Student 2 - Answer to Q1

TCP uses a congestion window called cwnd to control how much data the sender can send. At the beginning it uses slow start where the window doubles every round trip time until it reaches the threshold. After the threshold, congestion avoidance begins and the window increases slowly by one segment every RTT. If a timeout occurs, the threshold is halved and the window becomes one segment, so slow start starts again. If three duplicate ACKs are received, TCP does fast retransmit and sends the missing packet again without waiting. Then it goes to fast recovery, where the window is reduced to half. This way TCP reduces congestion and shares the network fairly. The sender also respects the receiver window.